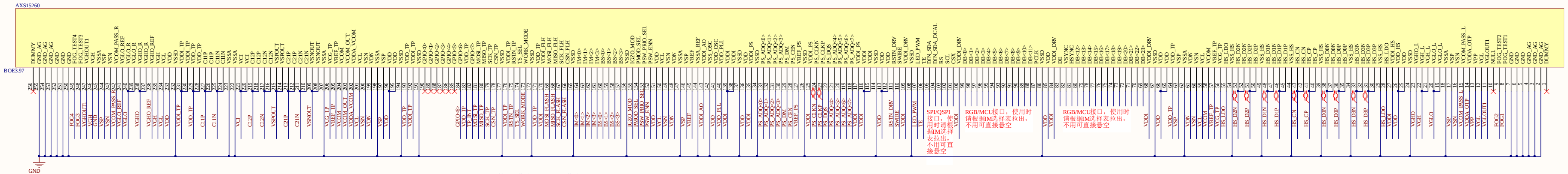
金手指

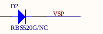

AXS15260
BOE3.97
GND
RGB/MCU接口，使用时请根据IM选择表拉出，不用可直接悬空
RGB/MCU接口，使用时请根据IM选择表拉出，不用可直接悬空

VGHOUT1/2、VGLOUT1/2：panel是a-si工艺用OUT1，LTPS工艺用OUT2
PSW_ENN is power exchanger set pin: 0:enable ;1:disable,Default=1
PMOD_SEL is power voltage set pin: 0:when VDDI=3.3v 1:when VDDI=1.8v,Default=1
WORK_MODE is Selection of tp work interface：1:IIC，0:SPI ,Default=0　补充SCK_TP复用IIC的SCL ,MOSI_TP复用IIC的SDA
PSW_PRIO_SEL：0:VDDI_TP优先，1:VDDI_DRV优先,Default=1
TS_SEL：测试pin
VPP：OTP电源，OTP时需外灌8.5V
ATEST：测试pin

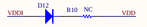
注意：（panel验证不用）

1.VSP上电比RST晚，导致不load flash问题，需在VDDI与VSP之间加二极管，VDDI电压起来后，将VSP电压带到1.8v

2.黑屏手势功能，在黑屏期间，需保持VDDI与VSP电压正常，TP_RST与DRV_RST为高电平

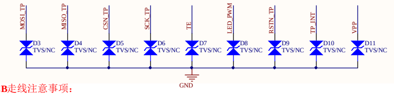

D2
VDDI
VSP
RBS520G/NC

FHD机型（panel验证不用）


D12
VDDI
R10
NC
VDD

注意：二极管型号和电阻阻值由实际情况决定

ESD

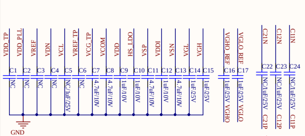

MOSI_TP
MISO_TP
CSN_TP
SCK_TP
TE
LED_PWM
RSTN_TP
TP_INT
VPP
D3 TVS/NC
D4 TVS/NC
D5 TVS/NC
D6 TVS/NC
D7 TVS/NC
D8 TVS/NC
D9 TVS/NC
D10 TVS/NC
D11 TVS/NC
GND

PCB走线注意事项：

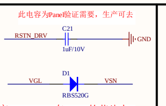
1、信号线布线尽可能的短，信号线太长时平行布一条地线，地线宽度信号线**2**倍宽度以上。
2、信号线和相应回路之间的环路面积尽可能小。布线比较长的信号线间隔一定距离和地线交换位置来减小环路面积。
3、避免在排线边缘放置重要的信号线，比如时钟和复位等。如果允许，所有元器件和布线都远离边缘。
4、排线上的测试点预留但不开窗，正反面都加上电磁屏蔽膜，屏蔽膜要接地良好。
5、提高排线的接地性能，增加排线的接地。

VDD_TP和VDD断开时，VDD_TP需要增加对地电容1uF/10V；

TP如果用全驱，VGLO和VGLO_REF、VGHO和VGHO_REF的电容需要加上；
TP如果用HIZ，这2个电容都得去掉。

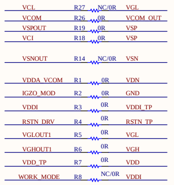

VDD_TP
VDD_PLL
VREF
VDN
VCL
VREF_TP
VCG_TP
VCOM
VDD
HS_LDO
VSP
VDDI
VSN
VGL
VGH
VGHO_REF
VGLO_REF
C21N
C12N
C11N
C1 NC
C2 NC
C3 NC
C4 NC
C5 NC/1uF/25V
C6 NC
C7 4.7uF/10V
C8 4.7uF/10V
C9 1uF/10V
C10 1uF/10V
C11 4.7uF/10V
C12 1uF/10V
C13 4.7uF/10V
C14 1uF/25V
C15 1uF/25V
C16 1uF/25V VGHO
C17 1uF/25V VGLO
C22 NC/1uF/25V C21P
C23 NC/1uF/25V C12P
C24 NC/1uF/25V C11P
GND

此电容为Panel验证需要，生产可去


C21
RSTN_DRV
1uF/10V
GND
D1
VGL
VSN
RBS520G

C7电容：VCG_TP在15260的芯片上需要贴上，在15261上是NC的；

C22、C23、C24电容：在15260芯片是不需要贴的，在15261芯片不贴VGH&VGL为内部升压，在15261芯片贴上VGH&VGL为外部电容升压


VCL R27 NC/0R VGL
VCOM R26 0R VCOM_OUT
VSPOUT R19 0R VSP
VCI R18 0R VSP
VSNOUT R14 NC/0R VSN
VDDA_VCOM R1 0R VDN
IGZO_MOD R2 0R GND
VDDI R3 0R VDDI_TP
RSTN_DRV R4 0R RSTN_TP
VGLOUT1 R5 0R VGL
VGHOUT1 R6 0R VGH
VDD_TP R7 0R VDD
WORK_MODE R8 NC/0R VDDI

电阻**R1**、**R5**、**R6**、**R18**、**R19**、**R26**：**15260**可去掉电阻并直连，**15261**验证后决定是否保留；

生产时**R8**的使用见**work_mode**描述；

电阻**R2**：玻璃验证完后决定是否接地；

R14是**15260**不贴，**15261**贴**0**欧电阻；

R27电阻在**15260**生产需要断开并删掉，**15261**保留；

注意：前期验证测试空贴位均不可去掉；

对于**HD**机型，**R7**可以**NC**，这时**VDD_TP**需要增加对地电容**1uF/10V**；
对于**FHD**机型，**R7**不能**NC**，这时**VDD_TP**可以不用电容

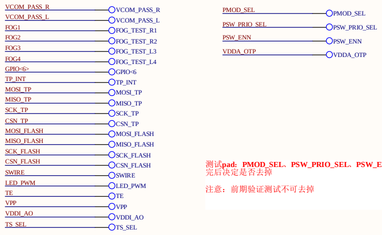
PCB走线注意事项：

1.MIPI走线需要按照标准差分走线要求，同时注意如果有过孔所有通道都要有相同的过孔，要么全过孔，要么全不过孔，不能有的通道过孔，有的通道不过孔；

2.TP_CLK和**TP_CSN**需要两边包地与其他线隔离开，防止干扰；如果条件允许**SPI**的走线都建议包地

3.测试焊盘按参考原理图预留；

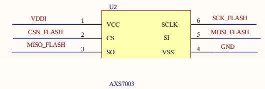
4.手机的**TP_IOVCC**和**TP_VCI**需要引焊盘出来；


VCOM_PASS_R
VCOM_PASS_L
FOG1 FOG_TEST_R1
FOG2 FOG_TEST_R2
FOG3 FOG_TEST_L3
FOG4 FOG_TEST_L4
GPIO<6>
TP_INT
MOSI_TP
MISO_TP
SCK_TP
CSN_TP
MOSI_FLASH
MISO_FLASH
SCK_FLASH
CSN_FLASH
SWIRE
LED_PWM
TE
VPP
VDDI_AO
TS_SEL
PMOD_SEL
PSW_PRIO_SEL
PSW_ENN
VDDA_OTP

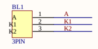
测试**pad**：**PMOD_SEL**、**PSW_PRIO_SEL**、**PSW_ENN**验证完后决定是否去掉

注意：前期验证测试不可去掉

FLASH


U2
VDDI
CSN_FLASH
MISO_FLASH
VCC
CS
SO
SCLK
SI
VSS
SCK_FLASH
MOSI_FLASH
GND
AXS7003

背光


BL1
A
K1
K2
3PIN

1.IM<3:0>内置上拉电阻，不需要外部上拉
2.不用的接口，皆可悬空！

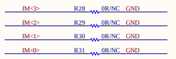
不支持3WSPI+RGB

外部连接IM[3:0]	Interface	Connect Pin
1111	MIPI	HS_DN,HS_DP,HS_CN,HS_CP,RSTN
1110	RGB 16/18/24-bit	DB[15/17/23:0],VSYNC_QSPI_DIN2,HSYNC,DE_QSPI_DIN3,PCLK
1101	SPI 4wire/8bits RGB	DIN_SDA(MISO),DIN_SDA_DUAL(MOSI),RS(D/CX),SCL,CSX,RSTN
1100	spi 3wire 9bits	DIN_SDA(MISO),DIN_SDA_DUAL(MOSI),SCL,CSX,RSTN
1011	SPI 4wire 8bits dual data lane	DIN_SDA,DIN_SDA_DUAL,RS(D/CX),SCL,CSX,RSTN
1010	MCU 8080 8bits	DB[7:0],VSYNC_QSPI_DIN2(CSX),HSYNC(D/CX),DE_QSPI_DIN3(WRX),PCLK(RDX),RSTN
1001	MCU 8080 9bits	DB[8:0],VSYNC_QSPI_DIN2(CSX),HSYNC(D/CX),DE_QSPI_DIN3(WRX),PCLK(RDX),RSTN
1000	MCU 8080 16bits	DB[15:0],VSYNC_QSPI_DIN2(CSX),HSYNC(D/CX),DE_QSPI_DIN4(WRX),PCLK(RDX),RSTN
0111	MCU 8080 18bits	DB[17:0],VSYNC_QSPI_DIN2(CSX),HSYNC(D/CX),DE_QSPI_DIN5(WRX),PCLK(RDX),RSTN
0110	SPI 3wire 9 bits dual lane	DIN_SDA,DIN_SDA_DUAL,SCL,CSX,RSTN
0101	QSPI	DE_QSPI_DIN3,VSYNC_QSPI_DIN2,DIN_SDA,DIN_SDA_DUAL,SCL,CSX,RSTN
0010	MCU 8080 8bits modeII	DB[17:10],VSYNC_QSPI_DIN2(CSX),HSYNC(D/CX),DE_QSPI_DIN5(WRX),PCLK(RDX),RSTN
0001	MCU 8080 9bits modeII	DB[17:9],VSYNC_QSPI_DIN2(CSX),HSYNC(D/CX),DE_QSPI_DIN5(WRX),PCLK(RDX),RSTN

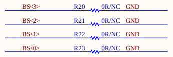
BS<3:0>内置上拉电阻，不需外部上拉

15260

外部连接BS[3:1]	外部连接BS[0]	HS_D3P	HS_D3N	HS_D0P	HS_D0N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D2P	HS_D2N
000	1	HS_D2P	HS_D2N	HS_D3P	HS_D3N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D0P	HS_D0N
001		HS_D0P	HS_D0N	HS_D3P	HS_D3N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D2P	HS_D2N
010		HS_D3P	HS_D3N	HS_D2P	HS_D2N	HS_CP	HS_CN	HS_D0P	HS_D0N	HS_D1P	HS_D1N
011		HS_D3P	HS_D3N	HS_D0P	HS_D0N	HS_CP	HS_CN	HS_D2P	HS_D2N	HS_D1P	HS_D1N
100		HS_D2P	HS_D2N	HS_D3P	HS_D3N	HS_CP	HS_CN	HS_D0P	HS_D0N	HS_D1P	HS_D1N
101		HS_D0P	HS_D0N	HS_D3P	HS_D3N	HS_CP	HS_CN	HS_D2P	HS_D2N	HS_D1P	HS_D1N
110		HS_D3P	HS_D3N	HS_D2P	HS_D2N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D0P	HS_D0N
111		HS_D3P	HS_D3N	HS_D0P	HS_D0N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D2P	HS_D2N
000	0	HS_D2N	HS_D2P	HS_D3N	HS_D3P	HS_CN	HS_CP	HS_D1N	HS_D1P	HS_D0N	HS_D0P
001		HS_D0N	HS_D0P	HS_D3N	HS_D3P	HS_CN	HS_CP	HS_D1N	HS_D1P	HS_D2N	HS_D2P
010		HS_D3N	HS_D3P	HS_D2N	HS_D2P	HS_CN	HS_CP	HS_D0N	HS_D0P	HS_D1N	HS_D1P
011		HS_D3N	HS_D3P	HS_D0N	HS_D0P	HS_CN	HS_CP	HS_D2N	HS_D2P	HS_D1N	HS_D1P
100		HS_D2N	HS_D2P	HS_D3N	HS_D3P	HS_CN	HS_CP	HS_D0N	HS_D0P	HS_D1N	HS_D1P
101		HS_D0N	HS_D0P	HS_D3N	HS_D3P	HS_CN	HS_CP	HS_D2N	HS_D2P	HS_D1N	HS_D1P
110		HS_D3N	HS_D3P	HS_D2N	HS_D2P	HS_CN	HS_CP	HS_D1N	HS_D1P	HS_D0N	HS_D0P
111		HS_D3N	HS_D3P	HS_D0N	HS_D0P	HS_CN	HS_CP	HS_D1N	HS_D1P	HS_D2N	HS_D2P


IM<3> R28 0R/NC GND
IM<2> R29 0R/NC GND
IM<1> R30 0R/NC GND
IM<0> R31 0R/NC GND

使用时请根据IM选择表拉出，内置上拉电阻，选1时可直接悬空，选0接GND


BS<3> R20 0R/NC GND
BS<2> R21 0R/NC GND
BS<1> R22 0R/NC GND
BS<0> R23 0R/NC GND

使用时请根据BS选择表拉出，内置上拉电阻，选1时可直接悬空，选0接GND

15261 BS选择

当(寄存器)cr_bs_opt=1时（默认），BS[3:0]由PAD决定，BS[4]由寄存器cr_bs_bit4配置

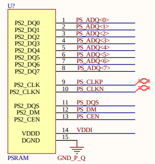
当(寄存器)cr_bs_opt=0时，BS[4:0]全部由寄存器cr_bs[4:0]配置

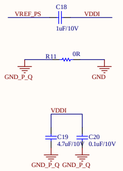
使用时请根据BS选择表拉出，内置上拉电阻，选1时可直接悬空，选0接GND

左侧为MIPI C-phy接口
右侧为MIPI D-phy接口

BS[4:1]/IC	BS[0]/IC	HS D2P	HS D2N	HS D1P	HS D1N	HS CP	HS CN	HS D0P	HS D0N	HS D3P	HS D3N	
4'b0000	1	HS_D2P	HS_D2N	HS_D3P	HS_D3N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D0P	HS_D0N	
4'b0001		HS_D0P	HS_D0N	HS_D3P	HS_D3N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D2P	HS_D2N	
4'b0010		HS_D3P	HS_D3N	HS_D2P	HS_D2N	HS_CP	HS_CN	HS_D0P	HS_D0N	HS_D1P	HS_D1N	
4'b0011		HS_D3P	HS_D3N	HS_D0P	HS_D0N	HS_CP	HS_CN	HS_D2P	HS_D2N	HS_D1P	HS_D1N	
4'b0100		HS_D2P	HS_D2N	HS_D3P	HS_D3N	HS_CP	HS_CN	HS_D0P	HS_D0N	HS_D1P	HS_D1N	
4'b0101		HS_D0P	HS_D0N	HS_D3P	HS_D3N	HS_CP	HS_CN	HS_D2P	HS_D2N	HS_D1P	HS_D1N	
4'b0110		HS_D3P	HS_D3N	HS_D2P	HS_D2N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D0P	HS_D0N	
4'b0111		HS_D3P	HS_D3N	HS_D0P	HS_D0N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D2P	HS_D2N	default
4'b1001		HS_D0P	HS_D0N	HS_D1P	HS_D1N	HS_CP	HS_CN	HS_D2P	HS_D2N	HS_D3P	HS_D3N	
4'b1100		HS_D2P	HS_D2N	HS_D1P	HS_D1N	HS_CP	HS_CN	HS_D0P	HS_D0N	HS_D3P	HS_D3N	
4'b0000	0	HS_D2N	HS_D2P	HS_D3N	HS_D3P	HS_CN	HS_CP	HS_D1N	HS_D1P	HS_D0N	HS_D0P	
4'b0001		HS_D0N	HS_D0P	HS_D3N	HS_D3P	HS_CN	HS_CP	HS_D1N	HS_D1P	HS_D2N	HS_D2P	
4'b0010		HS_D3N	HS_D3P	HS_D2N	HS_D2P	HS_CN	HS_CP	HS_D0N	HS_D0P	HS_D1N	HS_D1P	
4'b0011		HS_D3N	HS_D3P	HS_D0N	HS_D0P	HS_CN	HS_CP	HS_D2N	HS_D2P	HS_D1N	HS_D1P	
4'b0100		HS_D2N	HS_D2P	HS_D3N	HS_D3P	HS_CN	HS_CP	HS_D0N	HS_D0P	HS_D1N	HS_D1P	
4'b0101		HS_D0N	HS_D0P	HS_D3N	HS_D3P	HS_CN	HS_CP	HS_D2N	HS_D2P	HS_D1N	HS_D1P	
4'b0110		HS_D3N	HS_D3P	HS_D2N	HS_D2P	HS_CN	HS_CP	HS_D1N	HS_D1P	HS_D0N	HS_D0P	
4'b0111		HS_D3N	HS_D3P	HS_D0N	HS_D0P	HS_CN	HS_CP	HS_D1N	HS_D1P	HS_D2N	HS_D2P	
4'b1001		HS_D0N	HS_D0P	HS_D1N	HS_D1P	HS_CN	HS_CP	HS_D2N	HS_D2P	HS_D3N	HS_D3P	
4'b1100		HS_D2N	HS_D2P	HS_D1N	HS_D1P	HS_CN	HS_CP	HS_D0N	HS_D0P	HS_D3N	HS_D3P	

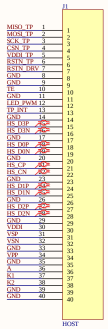
PSRAM（panel验证不用）


U7
PS2_DQ0
PS2_DQ1
PS2_DQ2
PS2_DQ3
PS2_DQ4
PS2_DQ5
PS2_DQ6
PS2_DQ7
PS2_CLK
PS2_CLKN
PS2_DQS
PS2_DM
PS2_CEN
VDDD
DGND
PS_ADQ<0>
PS_ADQ<1>
PS_ADQ<2>
PS_ADQ<3>
PS_ADQ<4>
PS_ADQ<5>
PS_ADQ<6>
PS_ADQ<7>
PS_CLKP
PS_CLKN
PS_DQS
PS_DM
PS_CEN
VDDI
PSRAM
GND_P_Q


VREF_PS
C18
1uF/10V
VDDI
R11
0R
GND_P_Q
GND
VDDI
C19
4.7uF/10V
C20
0.1uF/10V
GND_P_Q
GND_P_Q

1.PSRAM不用时，所有**pin**可以悬空
2.PSRAM不用时，需要关掉使能（使能默认关）
3.psram数据线尽量短，等长，间隔不要太近

接口


J1
MISO_TP
MOSI_TP
SCK_TP
CSN_TP
VDDI_TP
RSTN_TP
RSTN_DRV
GND
GND
TE
GND
LED_PWM
TP_INT
GND
HS_D3P
HS_D3N
GND
HS_D0P
HS_D0N
GND
HS_CP
HS_CN
GND
HS_D1P
HS_D1N
GND
HS_D2P
HS_D2N
GND
VDDI
VSP
VSN
GND
GND
VPP
A
K1
K2
GND
GND
HOST

Title
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